

High-chance portions - env-biodiversity

Combined original study articles for this topic's highest-probability PYQ themes.

Protected Areas, WPA & Species

~86% · Protected area categories · WPA schedules · Species habitats · Biosphere zones · Tiger Reserves

Why this portion (PYQ lens)

Portion chance ~86% (stable-high). Theme counts across all 10 years; parent env-biodiversity ~85%.

Years appeared (synthesis): 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025

Why UPSC asks this

Map memory + legal categories + flagship species create unambiguous MCQs.

Syllabus framing

Portion chance ~86% (stable-high). Theme counts across all 10 years; parent env-biodiversity ~85%. This portion sits inside env-biodiversity and centres on protected area categories, WPA and species - habitat matching. Map memory + legal categories + flagship species create unambiguous MCQs.

Recurring PYQ angles: Protected area categories, WPA schedules, Species habitats, Biosphere zones, Tiger Reserves. Approximate appearance years in the 2016-2025 educational synthesis: 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025. Use the chance percentage only to sequence study hours - it is not an official prediction.

Core map of protected area categories, WPA and species - habitat matching

National Park: Higher protection intensity in textbook contrasts. Wildlife Sanctuary: Protection with comparatively more flexible rights/activities.

Conservation/Community Reserves: Community and landscape approaches under WPA narrative. Biosphere Reserve: Core/buffer/transition zoning - not identical to NP.

Hold this core map as your elimination spine before memorising peripheral CA overlays.

Core concepts and definitions

Tiger Reserve: Critical tiger habitat + buffer under NTCA overlay. WPA 1972: Principal wildlife statute; schedules grade species protection.

Schedules logic: Higher schedule generally means stricter prohibitions. Ramsar overlay: Wetlands of international importance often co-asked.

Institutions, processes and mechanisms

Corridors: Connectivity beyond island PAs. Invasive species: Ecological disruption examples.

Flagship species matching: GIB, dolphin, snow leopard, olive ridley - living list. NTCA: Governance hook for Project Tiger landscapes.

India angle and applied examples

Ex situ vs in situ: Zoos/gene banks vs habitat protection. Biodiversity Act: NBA/ABS outline level.

CITES: International trade permit logic. Human - wildlife conflict: Edge habitats and mitigation themes.

Comparisons, linkages and discriminators

Coastal links: Turtles/mangroves bridge topics. Critical Wildlife Habitat: FRA interface - headline awareness.

IUCN vs Indian law: Do not casually mix category systems. Manage distinctive sites: Not all 100+ NPs - uniqueness first.

Category discriminators

National Parks are treated in textbook contrasts as higher-intensity protected areas with stricter activity controls than typical Wildlife Sanctuaries. Conservation and Community Reserves recognise community and landscape approaches under the Wildlife Protection Act narrative. Biosphere Reserves use a core/buffer/transition zoning idea inspired by UNESCO thinking and are not legal synonyms of National Parks. Tiger Reserves combine critical tiger habitat and buffer under the NTCA/Project Tiger overlay and can overlap other PA categories.

Corridors matter because fragmentation turns protected areas into islands. In situ protection (habitat) contrasts with ex situ tools (zoos, gene banks). Ramsar wetland sites are often co-asked with PA maps.

India angle

Revise distinctive PAs/species; track Ramsar and schedule CA on a one-page sheet.

Question patterns and exam spine

Matching lists and multi-statement questions on protected area categories, WPA and species - habitat matching.

Revision spine: Definitions -> table -> India -> traps for protected area categories, WPA and species - habitat matching.

Eliminate anachronisms and category errors first, then test remaining options against table rows and must-remember triggers.

Trap watch-list (specific)

Avoid: BR identical to NP. Avoid: All sanctuaries ban all humans. Avoid: IUCN=Indian legal categories. Avoid: Outdated schedules. Avoid: Tiger buffer = BR transition.

Indian PA categories

Category | Core idea | Discriminator

National Park -- Strict PA -- Vs WLS intensity

Wildlife Sanctuary -- More flexibility historically -- Rights/activities

Biosphere Reserve -- Zoned landscape -- Not NP synonym

Tiger Reserve -- CTH + buffer -- NTCA hook

Map / list facts

- Western Ghats
- Sundarbans
- Kaziranga
- Desert NP/GIB
- River dolphin stretches

Common traps

- x BR identical to NP
- x All sanctuaries ban all humans
- x IUCN=Indian legal categories
- x Outdated schedules
- x Tiger buffer = BR transition

Must remember

- NP vs WLS vs BR
- WPA schedules idea
- Tiger Reserve CTH+buffer
- In situ vs ex situ
- Flagship -> habitat pairs
- Corridors

Quick revision checklist

- [] Reproduce table
- [] 5 must-remembers
- [] 3 traps
- [] India anchor
- [] 10 MCQs

Educational prioritisation heuristic - not an official UPSC prediction. (c) UPSC IAS Study Hub original notes.

Biodiversity Hotspots & Global Conventions

~72% · Hotspots · CBD · CITES · Nagoya · CMS

Why this portion (PYQ lens)

Portion chance ~72%. Steady; pairs with protected-areas. Parent ~85%.

Years appeared (synthesis): 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025

Why UPSC asks this

Definitional criteria + treaty purpose one-liners are easy to mark.

Syllabus framing

Portion chance ~72%. Steady; pairs with protected-areas. Parent ~85%. This portion sits inside env-biodiversity and centres on biodiversity hotspots and global environment conventions. Definitional criteria + treaty purpose one-liners are easy to mark.

Recurring PYQ angles: Hotspots, CBD, CITES, Nagoya, CMS. Approximate appearance years in the 2016-2025 educational synthesis: 2016, 2017, 2018, 2019, 2020, 2021, 2022, 2023, 2024, 2025. Use the chance percentage only to sequence study hours - it is not an official prediction.

Core map of biodiversity hotspots and global environment conventions

Hotspot criteria: High endemism + habitat loss/threat thresholds. India-related hotspots: Himalaya, Western Ghats, Indo-Burma, Sundaland (Nicobar).

Endemism: Species restricted to a region. CBD: Conservation, sustainable use, benefit sharing.

Hold this core map as your elimination spine before memorising peripheral CA overlays.

Core concepts and definitions

Nagoya Protocol: Access and benefit sharing. CITES: International trade regulation via appendices.

CMS: Migratory species cooperation. Cartagena Protocol: Biosafety/LMOs - distinct from Nagoya.

Institutions, processes and mechanisms

Protected area != hotspot: Scientific label vs legal tool. Benefit sharing: Genetic resources and TK ethics.

Appendices idea: Different trade strictness. Migratory flyways: Birds link CMS and wetlands.

India angle and applied examples

ABS India: Biodiversity Act outline. Marine biodiversity: Reefs/mangroves/EEZ.

Invasive alien species: CBD concern. Aichi to GBF: Targets evolve - verify CA labels.

Comparisons, linkages and discriminators

Community knowledge: PBR idea. Synthetic biology edge: Headline awareness only.

India angle

Western Ghats endemism examples + NBA outline.

Question patterns and exam spine

Hotspot definition; convention -> purpose match.

Revision spine: Criteria -> India list -> convention table -> ABS/biosafety split.

Eliminate anachronisms and category errors first, then test remaining options against table rows and must-remember triggers.

Trap watch-list (specific)

Avoid: Every forest is a hotspot unit. Avoid: Cartagena=Nagoya. Avoid: CITES bans all trade. Avoid: CBD=UNFCCC.

Convention purposes

Instrument | Purpose

CBD -- Biodiversity triad

Nagoya -- ABS

Cartagena -- Biosafety

CITES -- Trade

CMS -- Migratory

Map / list facts

- Western Ghats
- Eastern Himalaya
- Nicobar

Common traps

- x Every forest is a hotspot unit
- x Cartagena=Nagoya
- x CITES bans all trade
- x CBD=UNFCCC

Must remember

- Hotspot criteria
- India hotspot list
- CBD triad
- Nagoya != Cartagena
- CITES=trade

Quick revision checklist

- [] Criteria
- [] India list
- [] 5 conventions
- [] 2 traps
- [] 10 MCQs

Educational prioritisation heuristic - not an official UPSC prediction. (c) UPSC IAS Study Hub original notes.